

Bitwise differential trail through **QALQAN** found by SAT

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1 QALQAN

Maximal differential probability: 2^{-61}

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	d	1	d	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	d	1	d	1
KeyAddStart																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	d	1	d	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	d	1	d	1
QalqaRound																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	1
QalqaRound																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1
QalqaRound																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
QalqaRound																											
0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	8	0	0	0	0	8	0	8	0
0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	8	0	0	0	0	8	0	8	0

2 QalqanRound

Maximal differential probability: $2^{-17.0}$

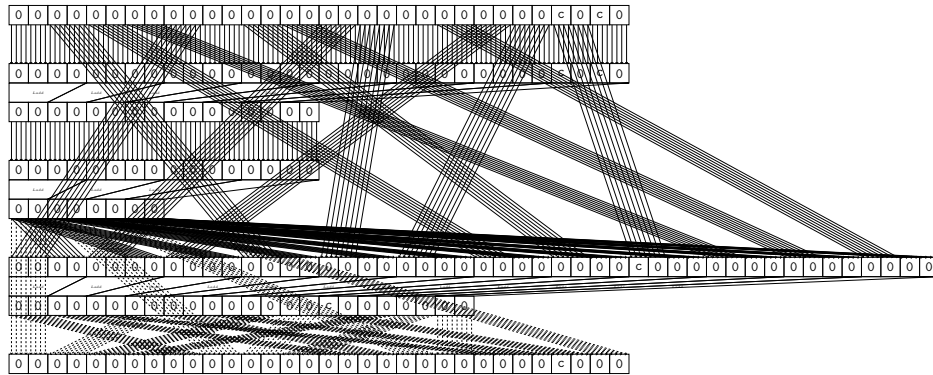
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	d	1	d	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	d	1	d	1
8 Blue Layers																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	c	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	c	0
L																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0
Add 126																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	1

3 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

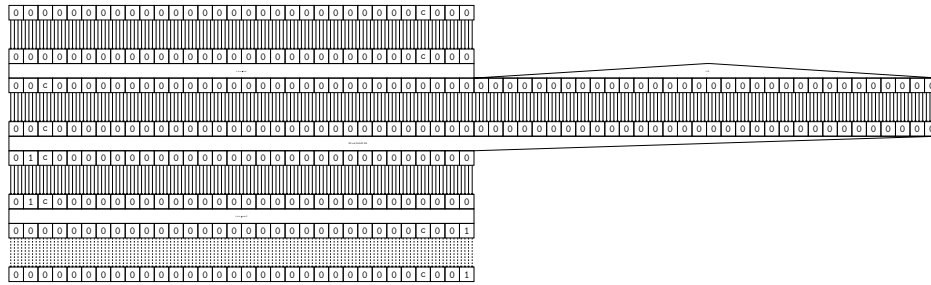
4 L

Maximal differential probability: $2^{-2.0}$



5 Add128

Maximal differential probability: $2^{-3.0}$



6 QalqanRound

Maximal differential probability: $2^{-16.0}$

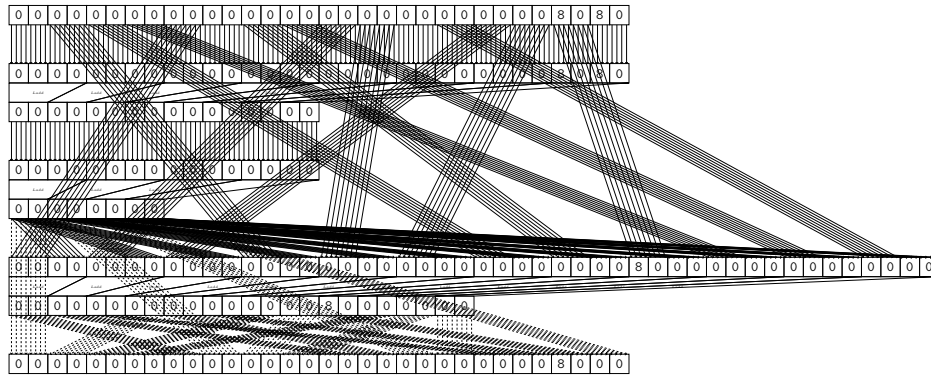
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	1
8 Blue Layers																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0
L																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0
Add 126																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1

7 SBoxLayer

Maximal differential probability: $2^{-14.0}$ [illegible]

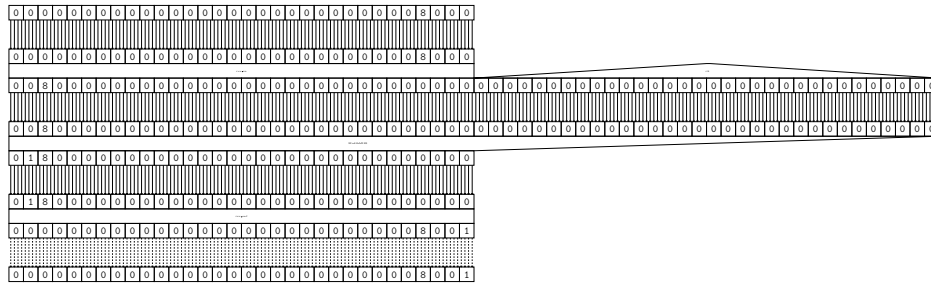
8 L

Maximal differential probability: $2^{-0.0}$



9 Add128

Maximal differential probability: $2^{-2.0}$



10 QalqanRound

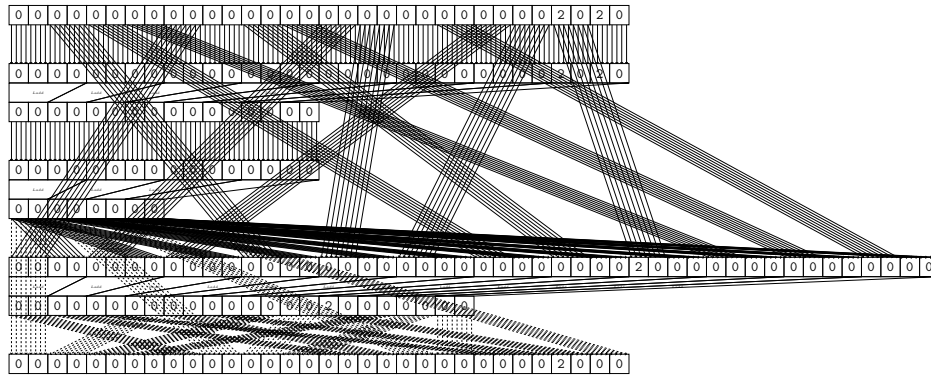
Maximal differential probability: $2^{-17.0}$ [illegible]

11 SBoxLayer

Maximal differential probability: $2^{-14.0}$ [illegible]

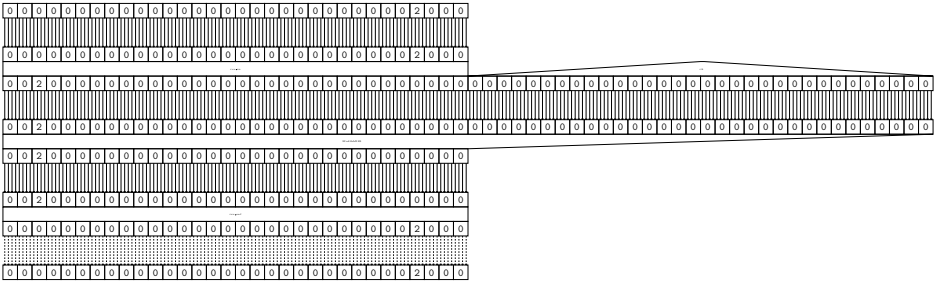
12 L

Maximal differential probability: $2^{-2.0}$



13 Add128

Maximal differential probability: $2^{-1.0}$



14 QalqanRound

Maximal differential probability: $2^{-11.0}$ [illegible]

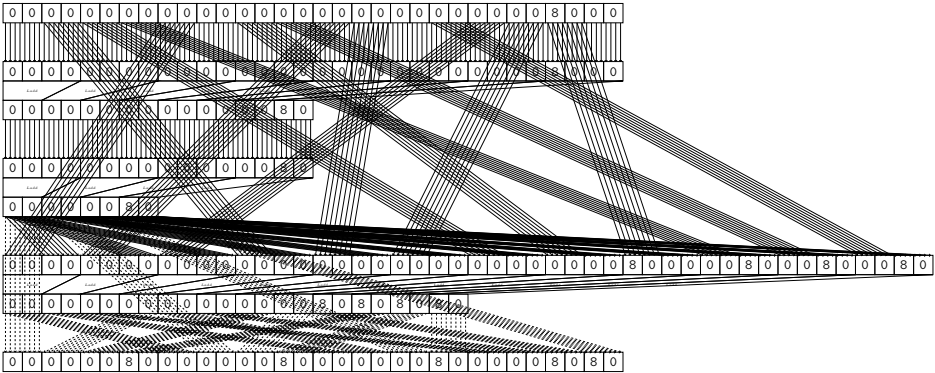
15 SBoxLayer

Maximal differential probability: $2^{-7.0}$

[illegible]

16 L

Maximal differential probability: $2^{-0.0}$



17 Add128

Maximal differential probability: $2^{-4.0}$

